

The Perils of Trivialising the IoT Security

This article is based on a talk by Sumanth Naropanth at the India Electronics Week 2019, organised by the EFY Group, where he spoke about who is responsible for securing IoT data.



Broadly, the Internet of Things (IoT) refers to a network of Internet-connected devices that are capable of gathering and sharing electronic information. The devices could be consumer goods, industrial equipment, traffic lights, construction equipment, healthcare devices and so on.

Data is the blood of IoT devices and, hence, it is important to manage and protect it. But how do you ensure integrity of data, its management or whether things are interconnected? These are some of the complexities and problems one may face from security as well as functional points of view.

There are a number of IoT protocols including ZigBee, Zwave, radio frequency (RF) and 4G, which means there are multiple options to choose

from. So the question arises, is having multiple standards way too many? Or, are too many standards an indication of having no standards that are good enough for security purposes?

The IoT ecosystem: an engineer's viewpoint

If you look at the IoT ecosystem from an engineer's perspective, there are four parts to it, namely, devices, mobile (which may or may not exist in some form factors), cloud and all networking aspects of it, including protocols.

Form factor devices (gateways, nodes, Edge, etc) are small devices of varying sizes. Depending on functionality, these might collect just sensor data and sensor values. Among these, gateways have a rich stack in the sense that these send all

collected information back to the cloud. However, architectures may vary depending on use-cases and hardware. In some cases, you will see cellphones acting as a conduit or channel.

For instance, in the case of a fitness tracker, you install an app on Android or iOS, create an account and then start pairing the device and pushing all data to the backend. On the cloud side, typically, you have Amazon Web Services (AWS), Google Cloud Platform (GCP) or other types of private cloud services. It is just an implementation like backend Web service, hosted on a server.

The cloud, the third component, has a much richer stack. This is where all data is processed, managed and presented in a way that is useful to customers and businesses.